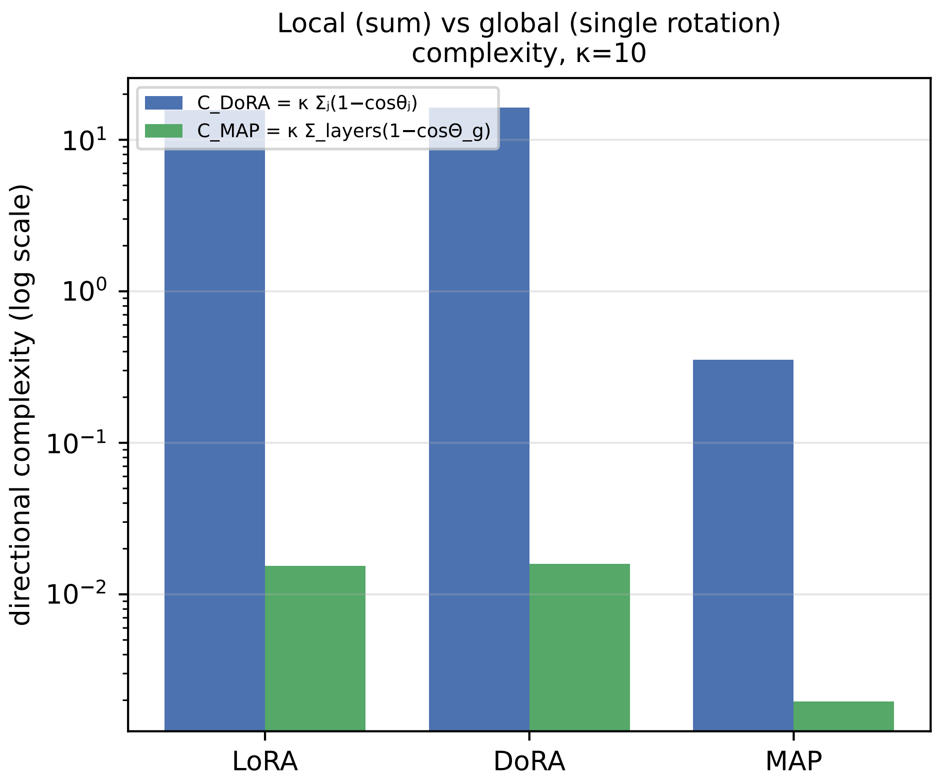
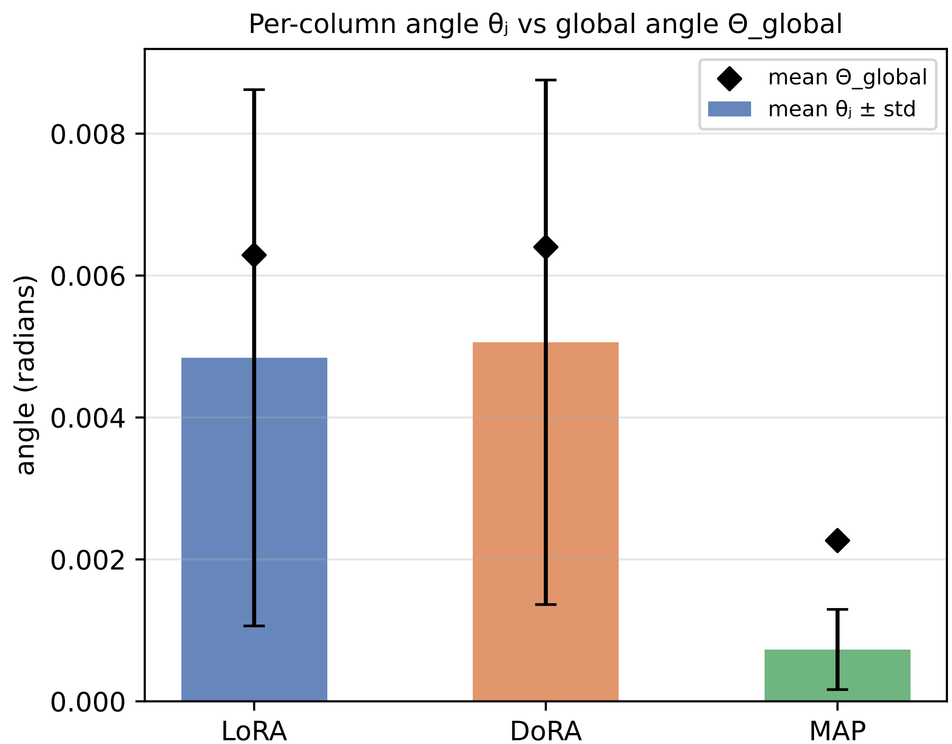
Directional Updates: LoRA vs DoRA vs MAP on RoBERTa-base / MRPC

Fine-tuning metrics and aggregate directional complexity

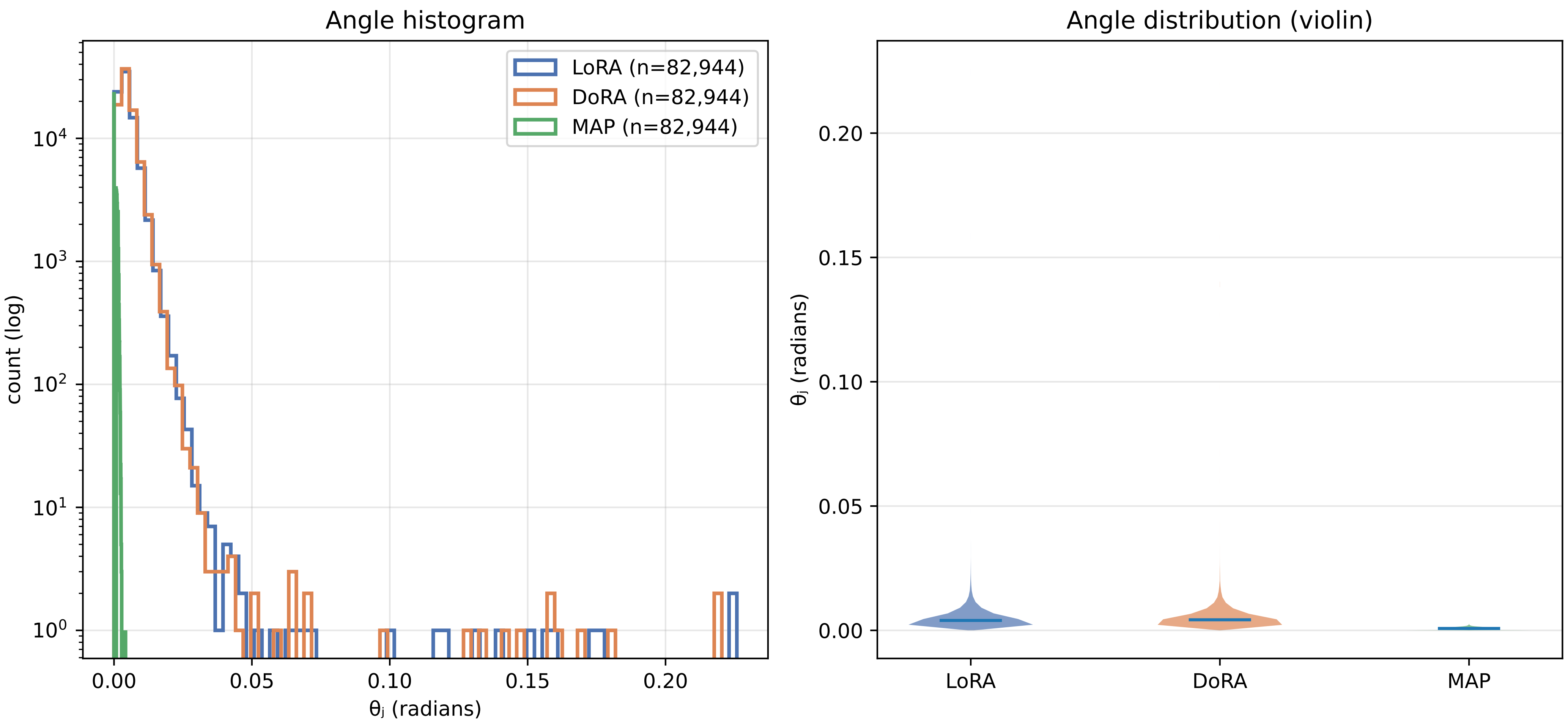
Method	Eval acc	Trainable params	C_DoRA (Σ local)	C_MAP (global)	mean θ_j (rad)	mean Θ_g (rad)	mean s/d
LoRA	84.07%	1,919,234	15.64	0.015	0.0048	0.0063	0.096
DoRA	85.70%	2,002,178	16.28	0.016	0.0051	0.0064	0.095
MAP	68.38%	1,919,378	0.35	0.002	0.0007	0.0023	0.000





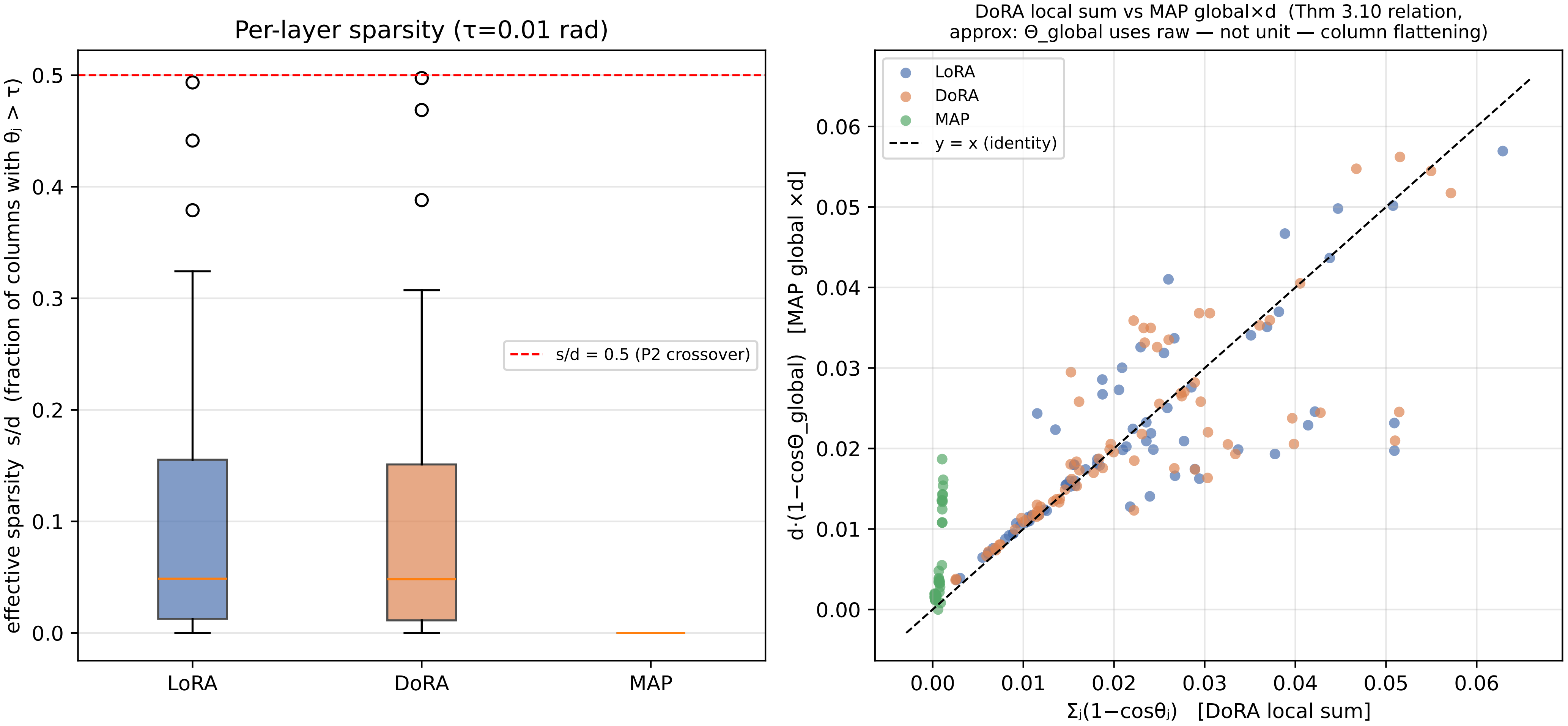
Θ_global is the single rotation of the whole flattened matrix (MAP view); θ_j are the per-column rotations (DoRA view).
Exact identity (Thm 3.10): $1-\cos\Theta\_global = mean\_j(1-\cos\theta_j)$ → DoRA sums what MAP averages.

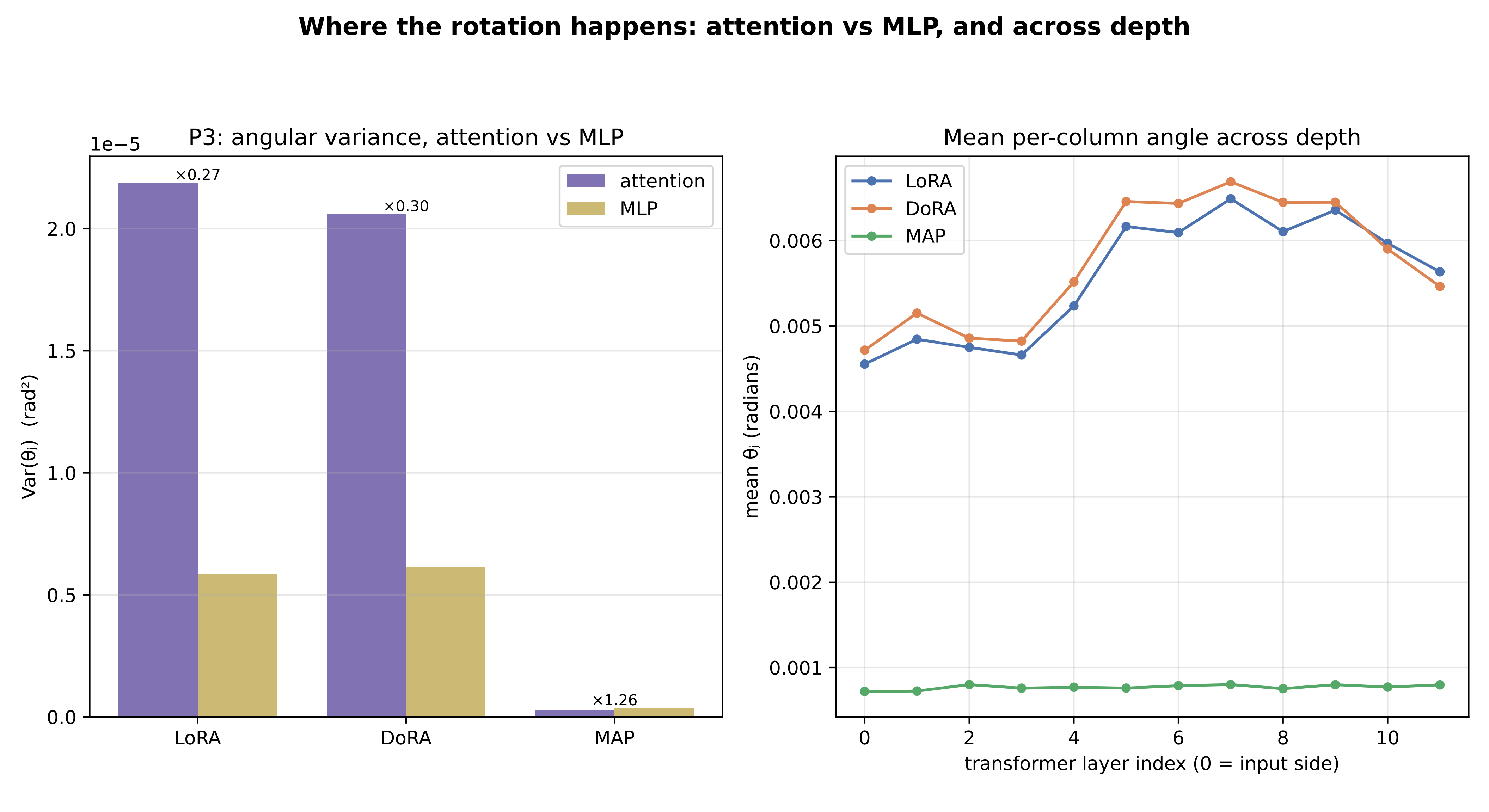
Distribution of per-column directional change θ_j (pooled over all target layers)



A long right tail = a few columns rotate a lot (sparse/localized update). A concentrated body near 0 = most columns barely move.

Sparsity of directional updates and the DoRA↔MAP identity





Where the rotation happens: attention vs MLP, and across depth
P3: angular variance, attention vs MLP
1e−5
×0.27
×0.30
×1.26
attention
MLP
Var(θj) (rad²)
LoRA
DoRA
MAP
Mean per-column angle across depth
mean θj (radians)
transformer layer index (0 = input side)